

The matrix dimension of a numerical semigroup $\langle 5, 7, 16, 18 \rangle$

4 October 2026

Theorem. For $S = \langle 5, 7, 16, 18 \rangle$,

$$\boxed{\dim_{\text{mat}} S = 3}.$$

Here $\mathcal{S}(A) = \{n \in \mathbb{N}_0 : A^n \in M_d(\mathbb{Z})\}$, and matricial dimension is the least possible d for which $\mathcal{S}(A) = S$.

This note gives an explicit rational matrix and a self-contained proof that its dimension is minimal. The accompanying Lean formalization proves the same result for every natural exponent, including both the upper and lower bounds. The construction can also be checked with exact rational arithmetic using the accompanying Python script.

1. Explicit matrix

Put

$$p = 11, \quad b = 833361690, \quad c = 2b - 1 = 1666723379.$$

Define

$$A = \begin{pmatrix} 0 & 0 & -c \, 11^8 \\ 11^{-13} & 0 & -b \, 11^{-6} \\ 0 & 11^8 & -11 \end{pmatrix}.$$

In decimal integers and fractions this is

$$A = \begin{pmatrix} 0 & 0 & -357276958458978899 \\ 1/34522712143931 & 0 & -833361690/1771561 \\ 0 & 214358881 & -11 \end{pmatrix}.$$

We prove that $\mathcal{S}(A) = S$.

2. The key polynomial identity

Let

$$C = \begin{pmatrix} 0 & 0 & -c \\ 1 & 0 & -b \\ 0 & 1 & -1 \end{pmatrix}, \quad D = \text{diag}(1, p^{14}, p^7).$$

Then $A = pD^{-1}CD$. Define

$$\delta = b^2 + b - 1, \quad \kappa = c(b - 1).$$

The chosen integer b satisfies

$$\delta = 294532279 \cdot 11^9, \quad b \equiv 7 \pmod{11}, \quad \kappa \equiv 1 \pmod{11}.$$

The characteristic polynomial of C is $f(x) = x^3 + x^2 + bx + c$. Polynomial reduction gives

$$x^5 \equiv c(b - 1) + (b^2 - b + c)x + (-1 + 2b - c)x^2 \pmod{f(x)}.$$

Because $c = 2b - 1$, Cayley-Hamilton yields the **exact** identity

$$\boxed{C^5 = \kappa I + \delta C}.$$

In particular, $C^5 \equiv \kappa I \pmod{11^9}$.

3. All required powers are integral

For every $n \geq 0$,

$$A^n = p^n D^{-1} C^n D.$$

Writing $(d_1, d_2, d_3) = (0, 14, 7)$, its (i, j) entry is

$$p^{n+d_j-d_i} (C^n)_{ij}.$$

Since C has integer entries and $d_i - d_j \leq 14$, **every** A^n **with** $n \geq 14$ **is integral**.

The fifth power is integral because

$$A^5 = p^5 \kappa I + p^5 \delta D^{-1} C D.$$

Every entry of the second term contains a factor whose exponent is at least $5 + 9 - 14 = 0$.

For the seventh power, use

$$C^7 = \kappa C^2 + \delta C^3.$$

The contribution $p^7 \delta D^{-1} C^3 D$ is integral, since its entrywise exponents are at least $7 + 9 - 14 = 2$. In the other contribution, $p^7 \kappa D^{-1} C^2 D$, the only potentially negative exponent occurs at position $(2, 1)$. But $(C^2)_{21} = 0$. Every other difference $d_i - d_j$ is at most 7. Hence A^7 is integral too.

Consequently $A^{10} = (A^5)^2$ and $A^{12} = A^5 A^7$ are integral.

The semigroup itself is

$$S = \{0, 5, 7, 10, 12\} \cup \{14, 15, 16, \dots\}.$$

Indeed, $14, 15, 16, 17, 18 \in S$, and adding the generator 5 gives every larger integer. We have therefore proved $S \subseteq \mathcal{S}(A)$.

4. Excluding every gap

It suffices to exclude 9, 11, 13, and then use closure of $\mathcal{S}(A)$.

In the entry $(2, 1)$, the conjugation formula becomes

$$(A^n)_{21} = p^{n-14} (C^n)_{21}.$$

From $C^5 = \kappa I + \delta C$, with $\delta \equiv 0 \pmod{p}$ and $\kappa \equiv 1 \pmod{p}$, we obtain

$$\begin{aligned} (C^9)_{21} &\equiv (C^4)_{21} = b - c = 1 - b \equiv 5 \pmod{11}, \\ (C^{11})_{21} &\equiv C_{21} = 1 \pmod{11}, \\ (C^{13})_{21} &\equiv (C^3)_{21} = -b \equiv 4 \pmod{11}. \end{aligned}$$

Thus these numerators are units modulo 11. The reduced denominators of the indicated entries are respectively $11^5, 11^3, 11$. In particular,

$$9, 11, 13 \notin \mathcal{S}(A).$$

For completeness, here is how those three exclusions exclude every remaining gap. If g were in $\mathcal{S}(A)$, then the displayed sum would be in $\mathcal{S}(A)$, since its second summand already belongs to $S \subseteq \mathcal{S}(A)$.

Gap g	Forbidden sum
1	$1 + 10 = 11$
2	$2 + 7 = 9$
3	$3 + 10 = 13$
4	$4 + 5 = 9$
6	$6 + 5 = 11$
8	$8 + 5 = 13$
9	Already excluded
11	Already excluded
13	Already excluded

These are all gaps of S . Therefore $\mathcal{S}(A) = S$, proving $\dim_{\text{mat}} S \leq 3$.

5. A self-contained lower bound

We prove that no rational matrix of dimension less than 3 can realize S . The key fact for dimension 2 is the following.

Lemma. If $M \in M_2(\mathbb{Q})$ and both M^5 and M^7 have integer entries, then M^{13} has integer entries.

Proof. Write $t = \text{tr } M$ and $d = \det M$. Since

$$d^5 = \det(M^5) \in \mathbb{Z},$$

the rational number d is an integer: a rational number in lowest terms whose fifth power is an integer has denominator 1.

Cayley-Hamilton gives

$$M^2 = tM - dI.$$

It follows that

$$\text{tr}(M^5) = t^5 - 5dt^3 + 5d^2t.$$

Put $z = \text{tr}(M^5) \in \mathbb{Z}$. Then t is a rational root of the monic polynomial

$$X^5 - 5dX^3 + 5d^2X - z \in \mathbb{Z}[X].$$

The rational-root theorem therefore gives $t \in \mathbb{Z}$.

For these t, d , define the integer coefficients

$$\begin{aligned} P &= d^4 - 18d^3t^2 + 7d^2t^4 + 6dt^6 - 2t^8, \\ Q &= t^4(-8d + 3t^2), \\ R &= 4d^6t + 2d^5t^3. \end{aligned}$$

We have the exact identity

$$\boxed{M^{13} = PM^5 + QM^7 + RI.}$$

Here is a direct way to verify it. Define $u_0 = 0$, $u_1 = 1$, and

$$u_{n+2} = tu_{n+1} - du_n.$$

Induction using Cayley-Hamilton gives

$$M^{n+1} = u_{n+1}M - du_nI \quad (n \geq 0).$$

Expanding the recurrence verifies the two polynomial identities

$$u_{13} = Pu_5 + Qu_7, \quad -du_{12} = -dPu_4 - dQu_6 + R.$$

Substitution proves the displayed matrix identity. Since its coefficients and all three matrices on its right are integral, M^{13} is integral. $\square$

If a 2×2 matrix realized S , its fifth and seventh powers would be integral because $5, 7 \in S$. The lemma would force its thirteenth power to be integral, contradicting $13 \notin S$.

In dimension 1, a realizing matrix is a rational scalar q . The condition $q^5 \in \mathbb{Z}$ forces $q \in \mathbb{Z}$, so exponent 1 would belong to its exponent semigroup. This contradicts $1 \notin S$. If dimension 0 is permitted, every power of the empty matrix is integral, which gives the same contradiction.

Thus $\dim_{\text{mat}} S \geq 3$. Together with the explicit 3×3 representation above, this proves

$$\dim_{\text{mat}} \langle 5, 7, 16, 18 \rangle = 3.$$

6. Exact verification and the choice of b

The verification script computes the least common denominator q_n of all entries of A^n . Its output is:

n	q_n
0	1
1	11^{13}
2	11^5
3	11^{11}
4	11^{10}
5	1
6	11^8
7	1
8	11^6
9	11^5
10	1
11	11^3
12	1
13	11
14-18	1

This finite verification extends to all exponents: the integral powers at 14 through 18, together with the integral fifth power, imply every power at least 14 is integral. All excluded exponents occur below 14.

The large coefficient b is deliberate. It is the lift of the root 7 of $x^2 + x - 1 \equiv 0 \pmod{11}$ to a root modulo 11^9 . The derivative $2 \cdot 7 + 1 \equiv 4 \pmod{11}$ is nonzero, so Hensel lifting applies. This produces the high divisibility in δ needed for the fifth power to become integral while the thirteenth power retains a denominator of 11.

7. Formal verification in Lean

The formalization uses ordinary Mathlib matrices over $\mathbb{Q}$. Integrality means that every entry is the cast of an integer, and membership in the target semigroup means

$$n = 5a + 7b + 16c + 18e \quad \text{for some } a, b, c, e \in \mathbb{N}_0.$$

The matrix dimension is defined as the least realizable dimension.

For the upper bound, Lean checks exact matrix-power certificates through exponent 18. It then proves the assertion for every exponent using induction and closure under multiplication. For the lower bound, it proves the 2×2 lemma in Section 5 and excludes dimensions 0 and 1.

The principal results in the source are:

Lean theorem	Mathematical statement
<code>witness_realizes</code>	The explicit matrix has exponent semigroup S .
<code>two_by_two_thirteenth_integral</code>	Integral fifth and seventh powers force an integral thirteenth power in dimension 2.
<code>matricial_dimension_eq_three</code>	The least matrix dimension is exactly 3.

The modular project and the complete single-file proof both compiled successfully with Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The axiom audit reports only Lean's standard `propext`, `Classical.choice`, and `Quot.sound`. The proof contains no `sorry`, extra axiom declarations, or uses of `native_decide`.

The accompanying project archive includes the Lean source, pinned dependencies, reproduction instructions, and the successful verification log.

References

1. Arsh Chhabra and Stephan Ramon Garcia, *Numerical semigroups from rational matrices III: semigroups of matricial dimension two and a counterexample to the lonely element conjecture*, Theorem 7.2 and Problem 8.2. arXiv:2506.11554.
2. Theo Chinn, Junshu Feng, Stephan Ramon Garcia, and Peiting Jiang, *Numerical semigroups from rational matrices IV: computation of the matricial dimensions of numerical semigroups with small Frobenius number or genus*, Problem 6.3 in the published version, *Bull. Aust. Math. Soc.* 114 (2026), 333-348. DOI:10.1017/S0004972726101063, arXiv:2510.15209.

Prepared 4 October 2026. This PDF includes the explicit construction and the independent lower-bound argument used in the verified Lean development. The references provide the surrounding literature and the published problem statement.